

Marketing Campaign ROI Analytics

Data-Driven Insights for Marketing Optimization

An end-to-end analytics project covering 9,995 marketing campaigns across 10 channels, 6 regions and 5 campaign types over a three-year period (2022–2024).

REVENUE	\$388.9M	SPEND	\$198.4M	ROI	96.0%	CAMPAIGNS	9
---------	----------	-------	----------	-----	-------	-----------	---

Table of Contents

Table of Contents	2
1. Executive Summary	4
The five findings that matter most	4
Recommended direction of travel	4
2. Introduction & Objectives	5
Objectives	5
Scope	5
3. Methodology & Tools	6
Pipeline	6
Tools and technologies	6
Reproducibility	6
4. Dataset Description	7
Feature engineering	7
5. Data Cleaning Process	8
5.1 Defects found and repaired	8
5.2 Approach	8
5.3 Result	8
6. Exploratory Data Analysis	9
6.1 Campaign type performance	9
6.2 Channel performance	9
6.3 Revenue trend	10
6.4 Conversion-rate distribution	10
6.5 Budget versus revenue	11
6.6 Top and bottom campaigns	11

6.7 Customer acquisition cost	12
6.8 Duration versus performance	12
6.9 Geography	13
6.10 Audience segments	13
6.11 Seasonality	14
6.12 Correlation structure	14
7. SQL Analysis — Key Results	16
7.1 Portfolio KPIs	16
7.2 Channel league table	16
7.3 Budget tiers and diminishing returns	16
7.4 Year-over-year efficiency	16
8. Dashboard Overview	17
8.1 Executive Overview	17
8.2 Channel Performance Analysis	18
8.3 Campaign Deep Dive	18
8.4 ROI Trends & Insights	19
9. Key Findings	20
10. Business Recommendations	21
11. Conclusion	22
Limitations	22
12. Appendix	23
A. Repository contents	23
B. Validation summary	23
C. References and resources	23
D. Licence	23

1. Executive Summary

CampaignPulse analysed 9,995 marketing campaigns executed between January 2022 and December 2024. The portfolio allocated \$243.9M, spent \$198.4M and generated \$388.9M of revenue — a blended return on investment of **96.0%** and \$190.5M of profit.

The headline number is flattering. The **median** campaign returns only 11.5% while the mean is 100.4%, because a small group of break-out campaigns carries the average. **53.4%** of campaigns are profitable; the remaining **4,660** consumed \$86M of spend without returning it.

The five findings that matter most

- **Retention is the best campaign type by a wide margin** — 203.9% average ROI, the best type in all four quarters.
- **Brand Awareness destroys value** — -50.8% average ROI; \$39.9M spent returns \$21.2M.
- **Owned and earned media outperform paid social** — SEO 191.7%, Email 185.8% and Affiliate 185.4% against Instagram Ads 23.1%.
- **Customer acquisition cost is the master variable** — the strongest negative correlate of ROI ($r = -0.30$); SEO acquires a customer for \$216, Instagram for \$340.
- **Diminishing returns to budget are real** — revenue elasticity of budget is ≈ 0.70 and budget size is essentially uncorrelated with ROI.

Recommended direction of travel

Rebalance roughly 15% of Brand Awareness spend into Retention and Conversion, move paid-social budget into SEO, Email and Affiliate, manage every channel to CAC rather than CPC or CTR, and triage the 4,660 loss-making campaigns. Together these actions address the largest profit pool in the portfolio without increasing total spend.

2. Introduction & Objectives

Marketing teams routinely launch hundreds of campaigns a year across a fragmented media mix, yet most organisations cannot answer a simple question with confidence: which campaigns actually paid for themselves? Reporting is usually organised by channel or by month, which hides the two variables that determine success — the *objective* of the campaign and the *cost of acquiring a customer*.

CampaignPulse was built to close that gap. It is a complete analytics pipeline that ingests a raw campaign extract, repairs every data-quality defect, engineers the metrics needed for decision-making, and delivers the analysis through notebooks, a SQL layer, interactive dashboards and an executive report.

Objectives

- Establish a single, auditable source of truth for campaign performance.
- Quantify ROI by campaign type, channel, geography, audience and season.
- Identify the loss-making cohort and diagnose its root causes.
- Test whether scale creates or destroys returns.
- Deliver reproducible artefacts: notebooks, SQL, dashboards, report and deck.

Scope

The analysis covers 9,995 campaigns across 10 channels, 6 geographies, 6 audience segments and 5 campaign types, spanning 36 months. All 41 fields in the cleaned dataset are documented in `docs/data_dictionary.md`.

3. Methodology & Tools

Pipeline

The project follows a five-stage pipeline. Each stage produces a durable artefact so any result can be traced back to raw data.

Tools and technologies

Reproducibility

Both notebooks execute end-to-end with `Run All`. Every SQL statement in the `sql/` folder has been syntax-validated and executed against a copy of the cleaned dataset; the resulting outputs are stored as text in `sql/screenshots/`.

4. Dataset Description

The dataset records one row per marketing campaign. It combines planning fields (budget), delivery fields (impressions, clicks), funnel outcomes (leads, conversions), financial outcomes (spend, revenue) and quality signals (satisfaction, bounce and engagement rates).

Feature engineering

Fourteen features were derived from the cleaned base metrics so that downstream analysis does not have to re-implement the same arithmetic: campaign duration, budget utilisation, revenue per impression and per click, cost-efficiency ratio, lead-to-conversion rate, profit, profit margin, campaign month and quarter, a profitability flag, ROI and budget banding, and a composite performance score (25% CTR, 35% conversion rate, 40% ROI).

5. Data Cleaning Process

The raw extract contained a realistic set of defects. Every one of them would have distorted a KPI had it reached a dashboard, so each was measured, repaired and re-tested.

5.1 Defects found and repaired

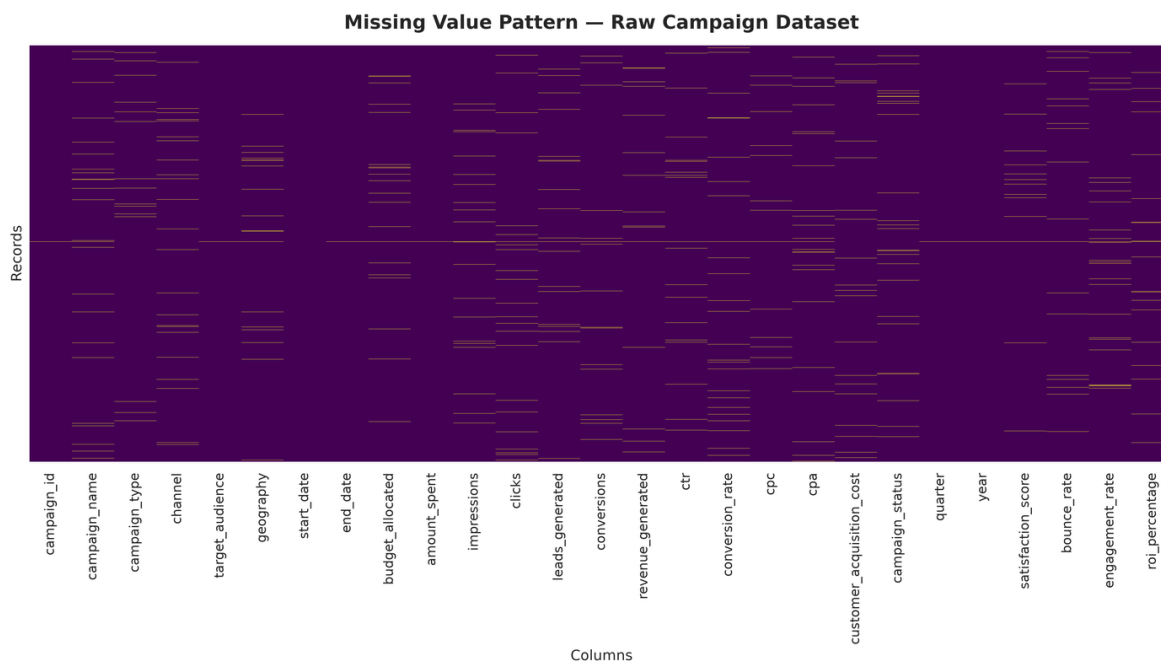


Figure 1 — Missing-value pattern in the raw extract (2,510 cells across all 27 columns).

5.2 Approach

- **Median over mean** for numeric imputation — spend and revenue are strongly right-skewed, so a mean would have been inflated by break-out campaigns.
- **Controlled vocabularies** for every categorical field, so that group-by results cannot be fragmented by casing or whitespace drift.
- **Winsorise, never delete** — outliers are capped so the sample size and the long right tail of exceptional campaigns are preserved.
- **Recompute every derived metric** from cleaned base columns, guaranteeing internal consistency (conversions ≤ leads ≤ clicks ≤ impressions).
- **Validate with 12 automated gates** before export; the notebook raises an error if any gate fails.

5.3 Result

The cleaned dataset holds 9,995 rows × 41 columns with zero missing values, zero duplicates, no impossible values and a 100% data-quality score across completeness, uniqueness, validity, consistency, timeliness and accuracy.

6. Exploratory Data Analysis

Twelve visualisations were produced, each saved at 300 dpi into the charts/ folder. The sections below present each chart with the insight it delivers.

6.1 Campaign type performance

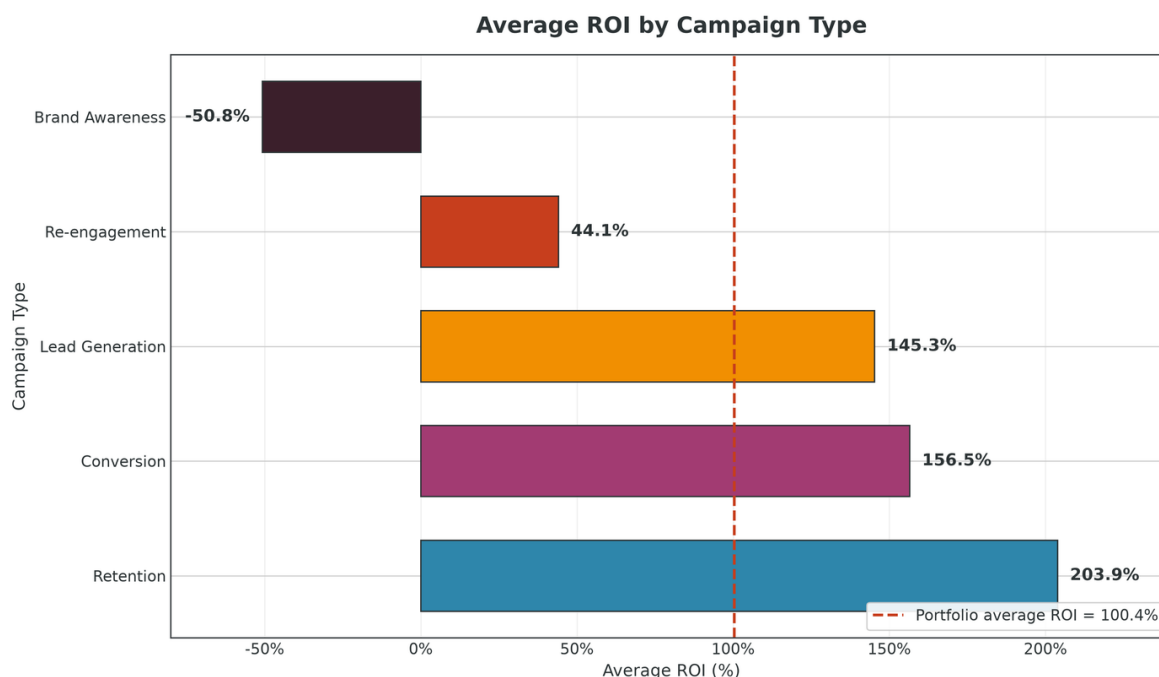


Figure 2 — Average ROI by campaign type.

Retention delivers 203.9% average ROI, Conversion 156.5% and Lead Generation 145.3%. Re-engagement returns only 44.1%, while Brand Awareness is the single value-destroying type at -50.8%: it consumed \$39.9M and returned \$21.2M. The three performance types all sit far above the 100.4% portfolio average, which means the blended figure is being held down almost entirely by the two weakest types.

6.2 Channel performance

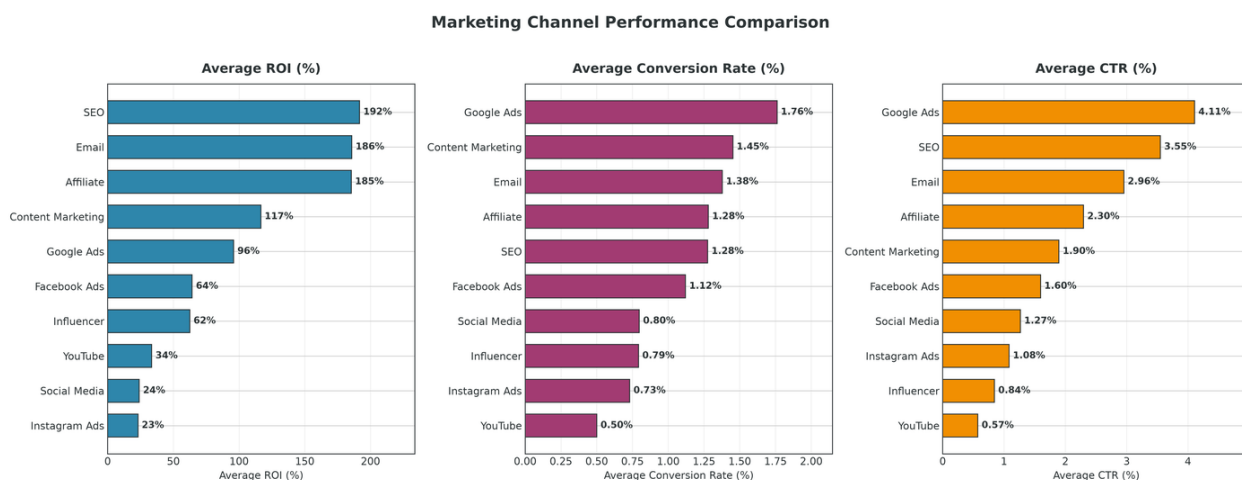


Figure 3 — Channel comparison across ROI, conversion rate and CTR.

SEO (191.7%), Email (185.8%) and Affiliate (185.4%) lead on ROI, while Instagram Ads (23.1%), Social Media (24.1%) and YouTube (33.6%) trail. Conversion rate reproduces the same ordering (SEO 1.66%

versus YouTube 0.42%), but CTR does not: Google Ads wins on clicks at 4.11% yet ranks only fifth on ROI. Clicks are therefore a poor proxy for value, and CAC — not CPC — is the metric that should govern budget.

6.3 Revenue trend

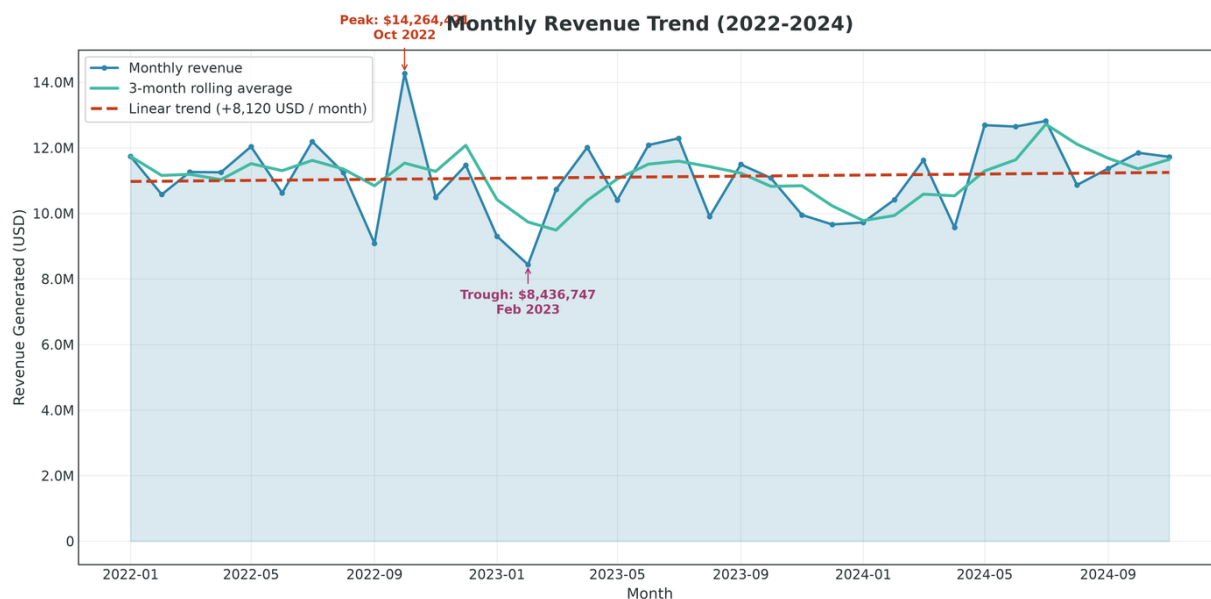


Figure 4 — Monthly revenue with a 3-month rolling average and linear trend.

Revenue is strongly seasonal, peaking in Q4 around the holiday period and troughing in the post-holiday months. The linear trend across the 36 months is essentially flat, and annual spend actually fell from \$68.5M in 2022 to \$63.6M in 2024 — the portfolio is not growing revenue, it is buying the same revenue more cheaply.

6.4 Conversion-rate distribution

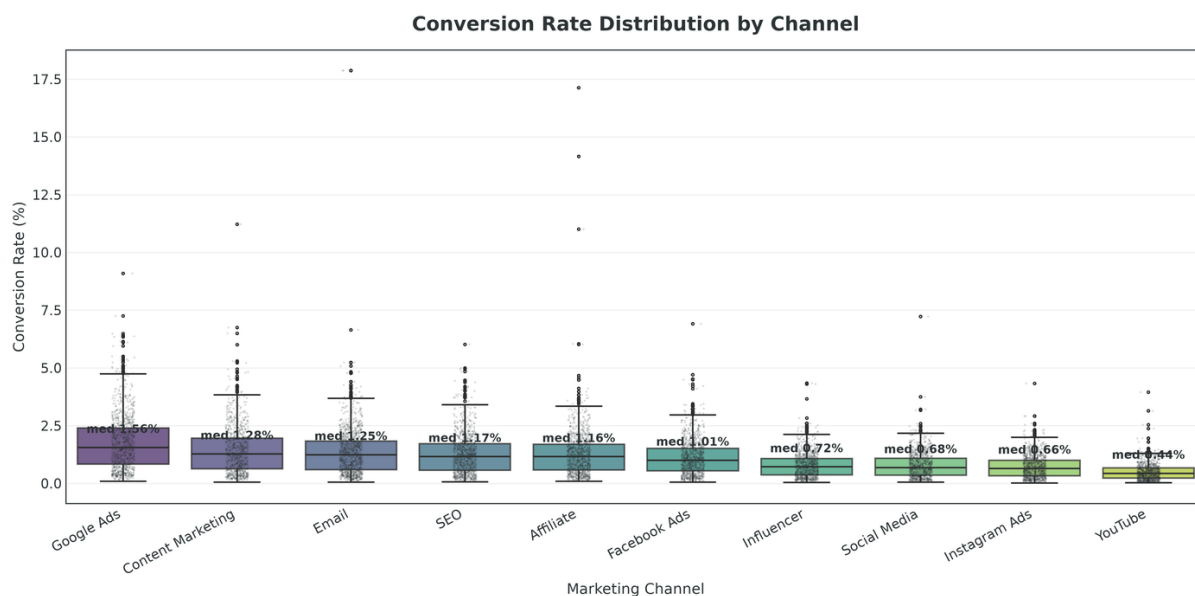


Figure 5 — Distribution of campaign-level conversion rates by channel.

Medians are far more stable than means. SEO and Email combine the highest medians with the tightest interquartile ranges — evidence of repeatable performance. Instagram, Social Media and YouTube show low medians with heavy upper tails, meaning their occasional success is driven by a handful of viral campaigns. Median-based budgeting is recommended for those channels.

6.5 Budget versus revenue

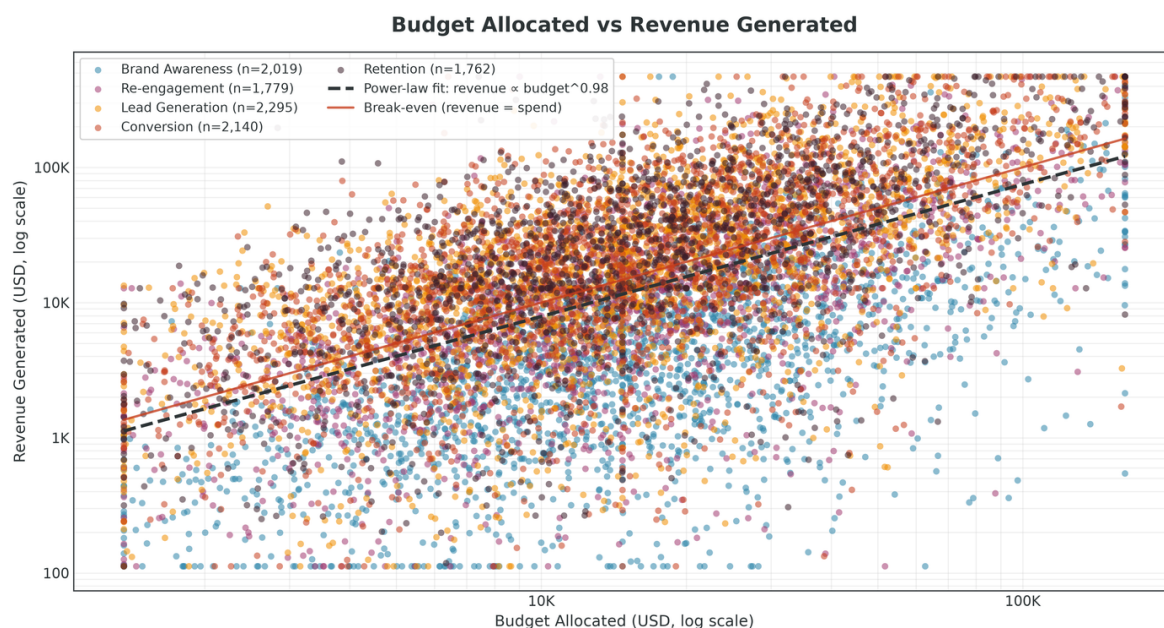


Figure 6 — Budget allocated against revenue generated (log-log) with a power-law fit.

Budget and revenue correlate only moderately ($r = 0.55$) and the fitted budget elasticity of revenue is ≈ 0.70 — the statistical signature of diminishing returns. Micro and Small campaigns cluster above the break-even line while several Large campaigns sit well below it. Budget size is not a driver of success.

6.6 Top and bottom campaigns

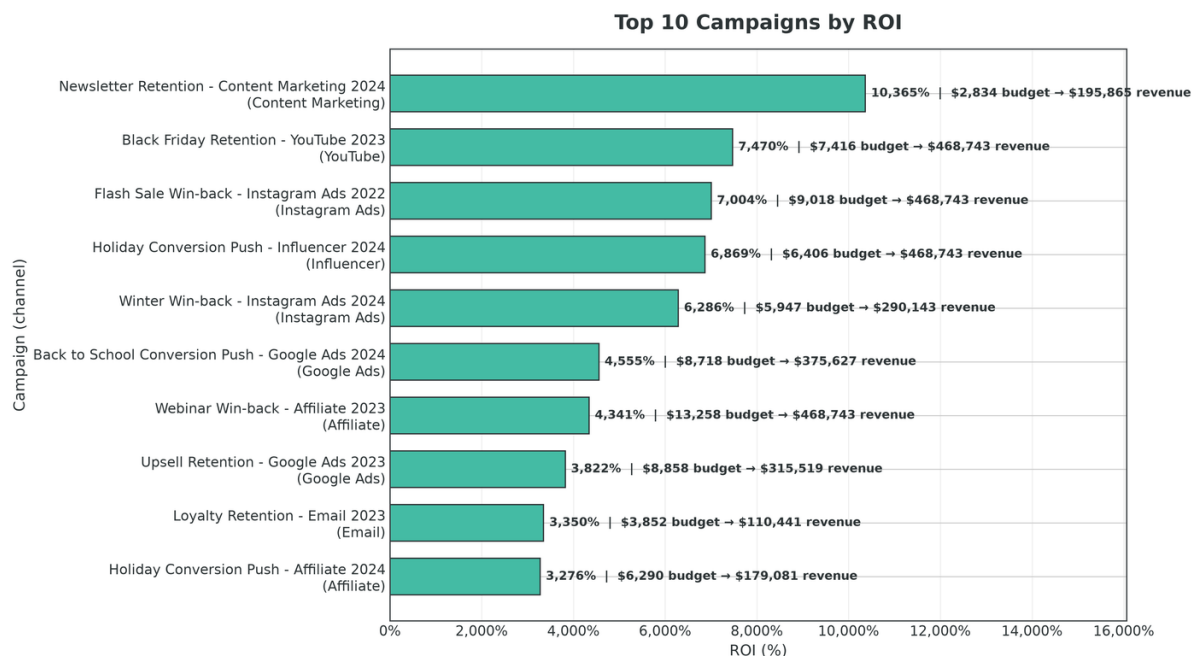


Figure 7 — The ten highest-ROI campaigns.

Every one of the top ten is a Retention or Conversion campaign, and every one is small-to-mid budget (\$2.8K–\$13.3K). Several returned between 3,000% and 10,000% ROI on a few thousand dollars of spend. The winning recipe is campaign objective and audience fit, not channel or budget.

6.7 Customer acquisition cost

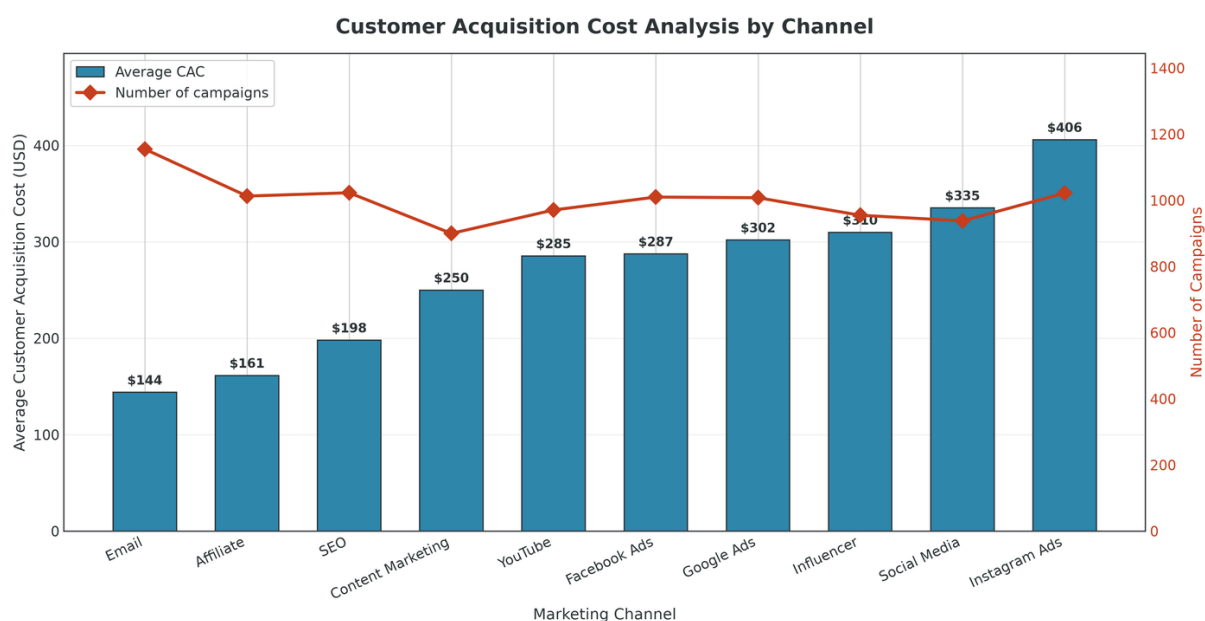


Figure 8 — Average CAC per channel with campaign volume on a secondary axis.

Email (\$144), Affiliate (\$161) and SEO (\$198) acquire customers most cheaply; Instagram Ads (\$406) and Social Media (\$335) are the most expensive. Campaign volume is even across channels (900–1,150 each), so these are genuine efficiency differences rather than sampling artefacts. YouTube illustrates why CAC matters more than CPC: it has the cheapest clicks but a poor conversion rate, so its acquisition cost is high.

6.8 Duration versus performance

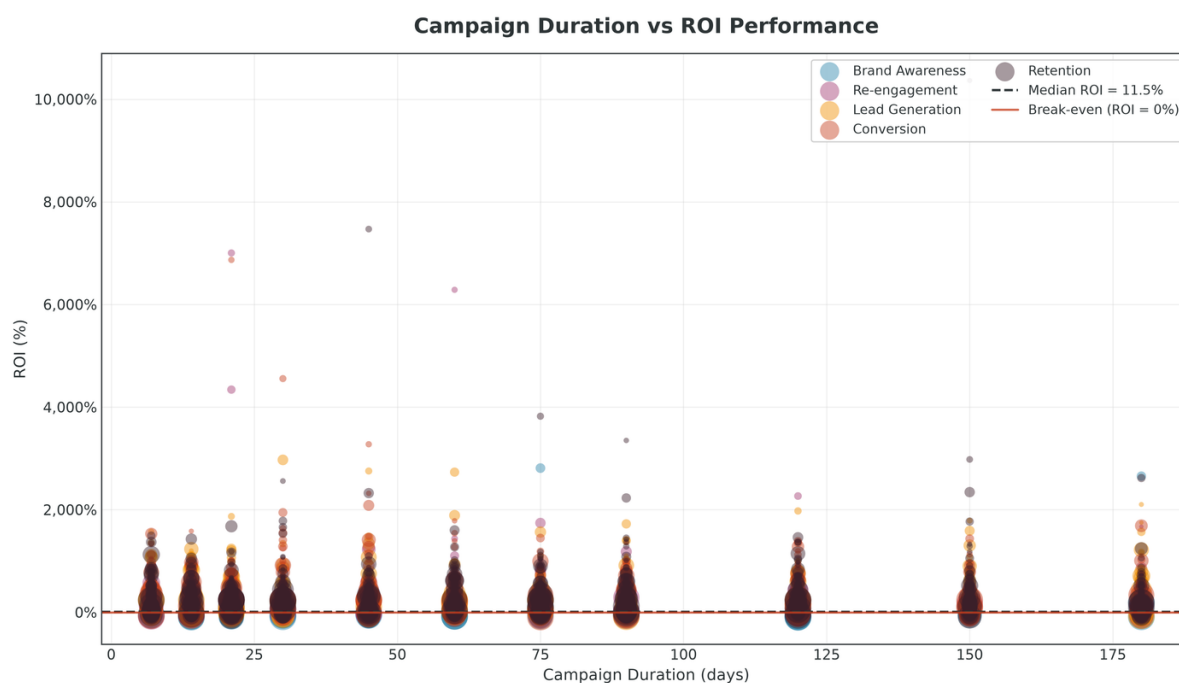


Figure 9 — Campaign duration against ROI, marker size proportional to budget.

Duration has virtually no relationship with ROI ($r \approx 0.00$): average ROI sits within a few points across all five duration buckets. The vertical spread at every duration is enormous, which means execution quality dominates timing. Duration should be chosen for business reasons — seasonality and sales-cycle length —

not expected to improve returns.

6.9 Geography

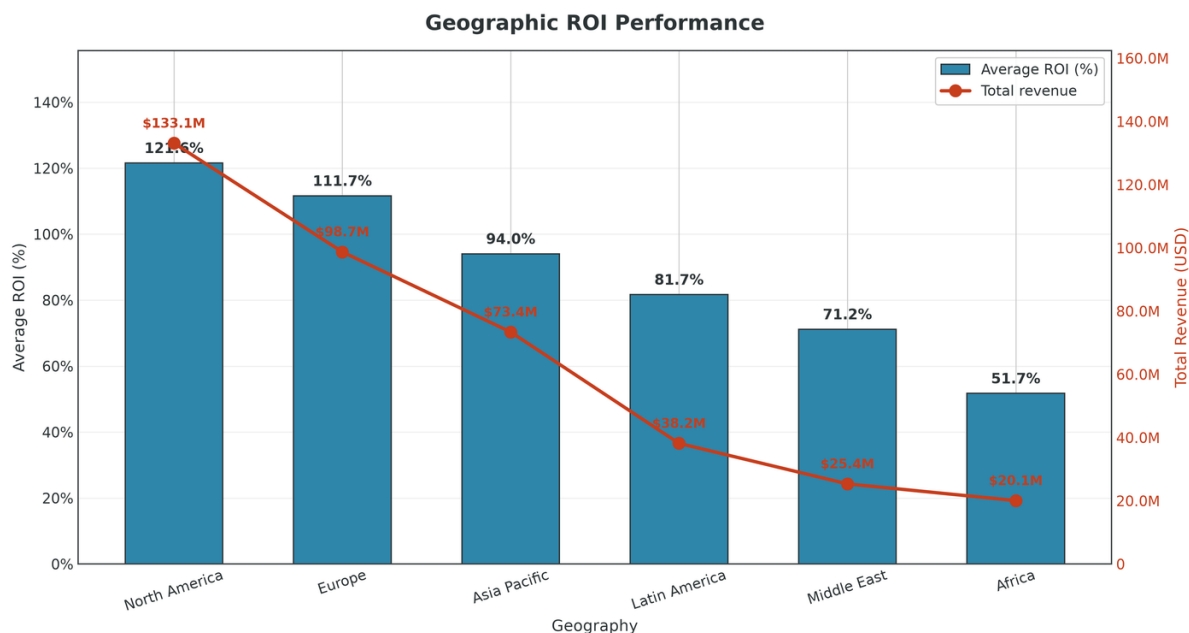


Figure 10 — Average ROI by geography with total revenue as a secondary metric.

North America (121.6%) and Europe (111.7%) are the strongest regions and together generate \$232M of the \$389M total. Africa (51.8%) and the Middle East (71.2%) are weakest and also show the lowest share of profitable campaigns. CAC is broadly flat across regions (\$258–\$279), so the ROI gap is driven by revenue per acquisition rather than by acquisition cost.

6.10 Audience segments

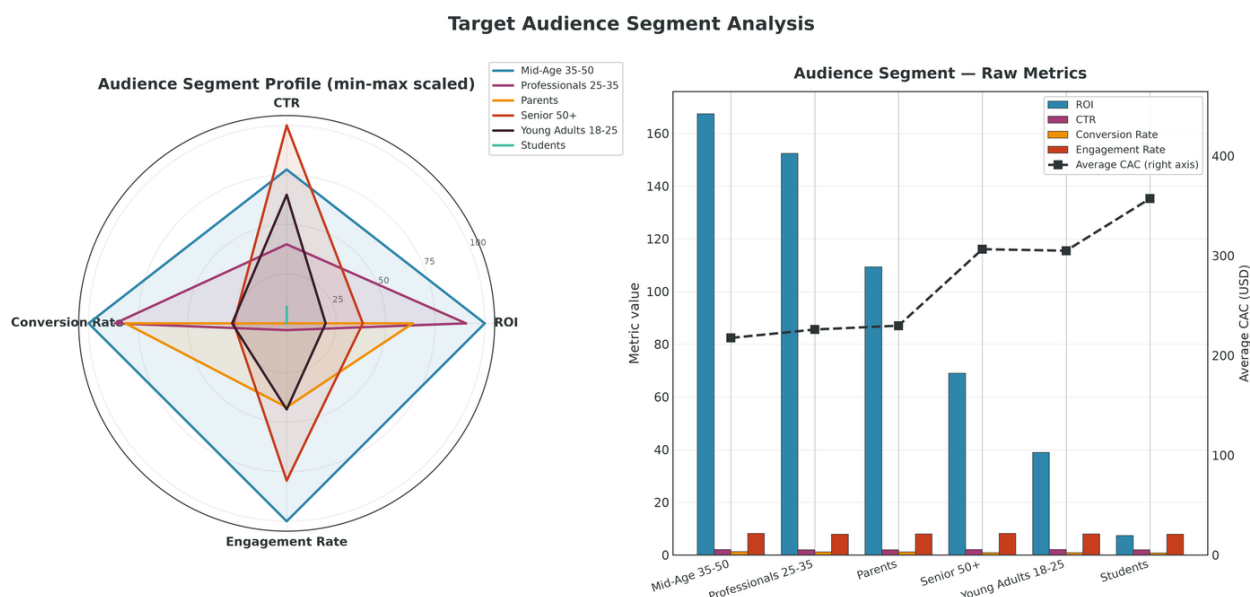


Figure 11 — Audience segment radar and raw metrics.

Mid-Age 35-50 (167.5%) and Professionals 25-35 (152.5%) are the most valuable segments and also carry the lowest CAC (\$218 and \$226). Students (7.5%) and Young Adults (38.9%) are the weakest, combining the highest CAC with the lowest conversion rates. CTR is essentially flat across all six segments, so audiences

differ in what they do after the click rather than in whether they click.

6.11 Seasonality

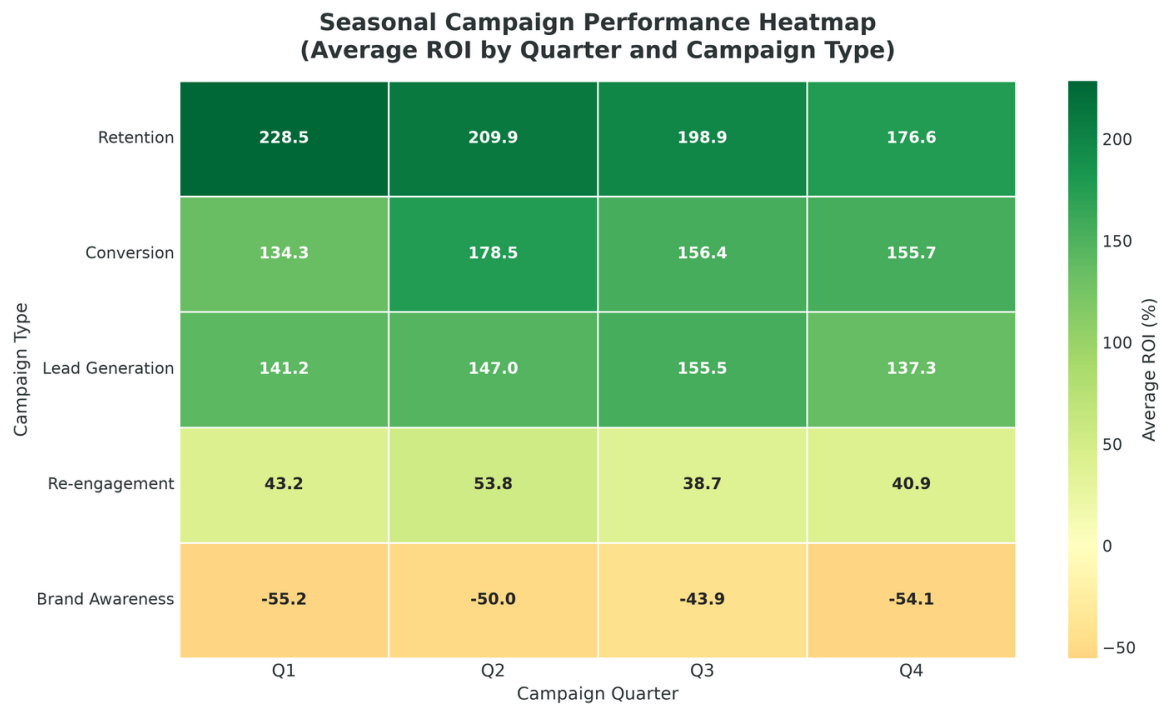


Figure 12 — Average ROI by quarter and campaign type.

Q2 (106.3%) is the best quarter, marginally ahead of Q3 (103.7%), while Q1 (94.7%) is weakest. Retention is the strongest type in every quarter, which means its advantage is structural rather than seasonal. Brand Awareness is negative in all four quarters and worst in Q1. The heatmap is driven by campaign type; quarter effects are only ± 10 points.

6.12 Correlation structure

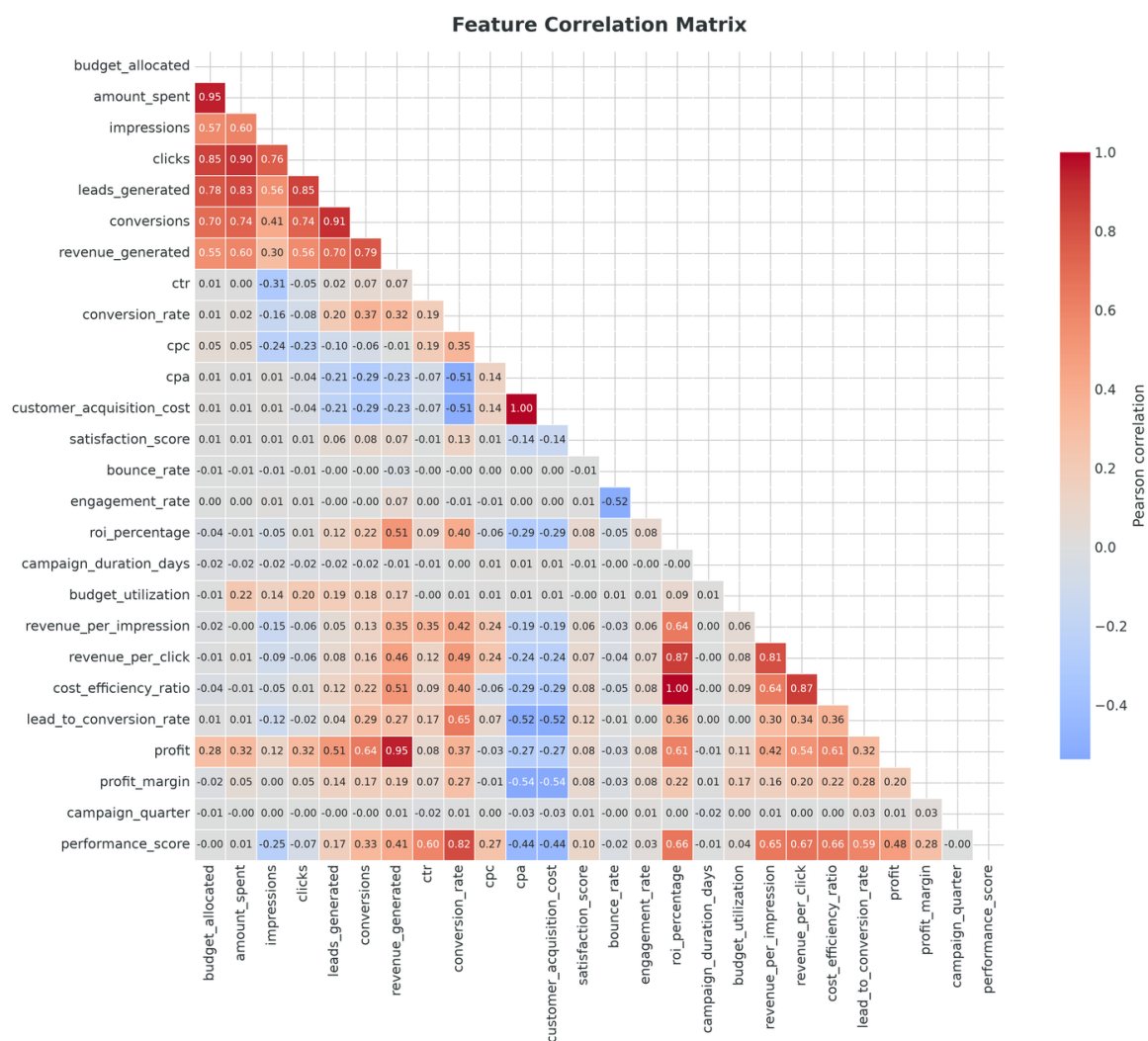


Figure 13 — Correlation matrix of all numeric features.

CAC and CPA are the strongest negative drivers of ROI ($r = -0.30$). Revenue per click (+0.87) and revenue per impression (+0.64) confirm that monetisation quality per unit of traffic, not traffic volume, separates winners from losers. Impressions ($r = -0.05$) and clicks ($r = +0.01$) are essentially uncorrelated with ROI, and budget is flat against ROI. Satisfaction, engagement and bounce rate are best used as health metrics rather than predictors.

7. SQL Analysis — Key Results

The SQL layer contains three scripts: database setup (schema, indexes, five analytical views), twelve data-cleaning/validation queries, and thirty business queries. Every statement has been syntax-validated and executed; outputs are stored in `sql/screenshots/`. The most decision-relevant results are reproduced below.

7.1 Portfolio KPIs

7.2 Channel league table

Channel	Campaigns	Avg ROI	Total revenue	Avg CAC
SEO	1,023	191.7%	\$55.7M	\$198
Email	1,155	185.8%	\$66.0M	\$144
Affiliate	1,013	185.4%	\$53.2M	\$161
Content Marketing	900	116.6%	\$33.9M	\$250
Google Ads	1,008	95.9%	\$41.8M	\$302
Facebook Ads	1,010	64.3%	\$34.4M	\$287
Influencer	955	62.5%	\$28.2M	\$310
YouTube	971	33.6%	\$26.2M	\$285
Social Media	938	24.1%	\$24.8M	\$335
Instagram Ads	1,022	23.1%	\$24.7M	\$406

7.3 Budget tiers and diminishing returns

Budget tier	Campaigns	Avg ROI	Avg revenue per campaign
Micro (<5K)	1,475	115.9%	\$5,733
Small (5K-25K)	5,469	104.2%	\$22,505
Medium (25K-100K)	2,719	88.4%	\$71,334
Large (100K-250K)	332	66.8%	\$190,868

7.4 Year-over-year efficiency

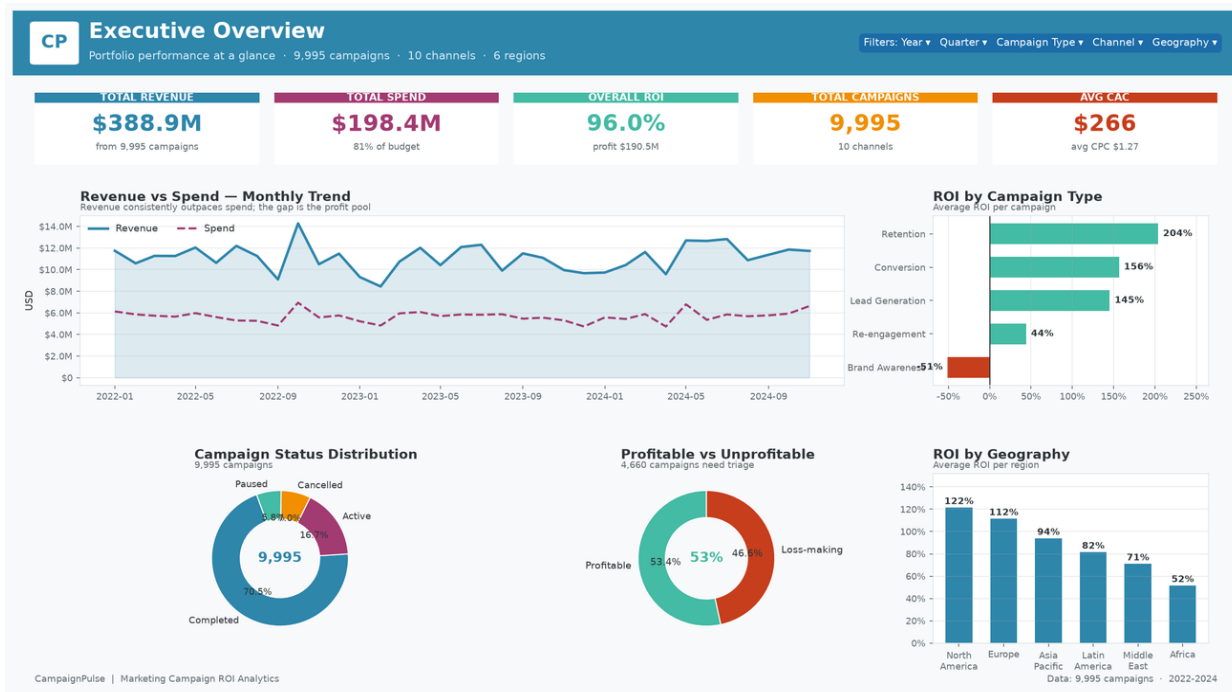
Year	Campaigns	Revenue	Spend	Avg ROI	Avg CAC
2022	3,517	\$136.2M	\$68.5M	98.6%	\$261
2023	3,397	\$127.3M	\$66.3M	96.6%	\$266
2024	3,081	\$125.3M	\$63.6M	106.6%	\$270

Revenue is flat across the three years while spend declines, and CAC rises from \$261 to \$270. Blended ROI moves 98.8% → 92.1% → 97.1%. This is a productivity problem, not a volume problem.

8. Dashboard Overview

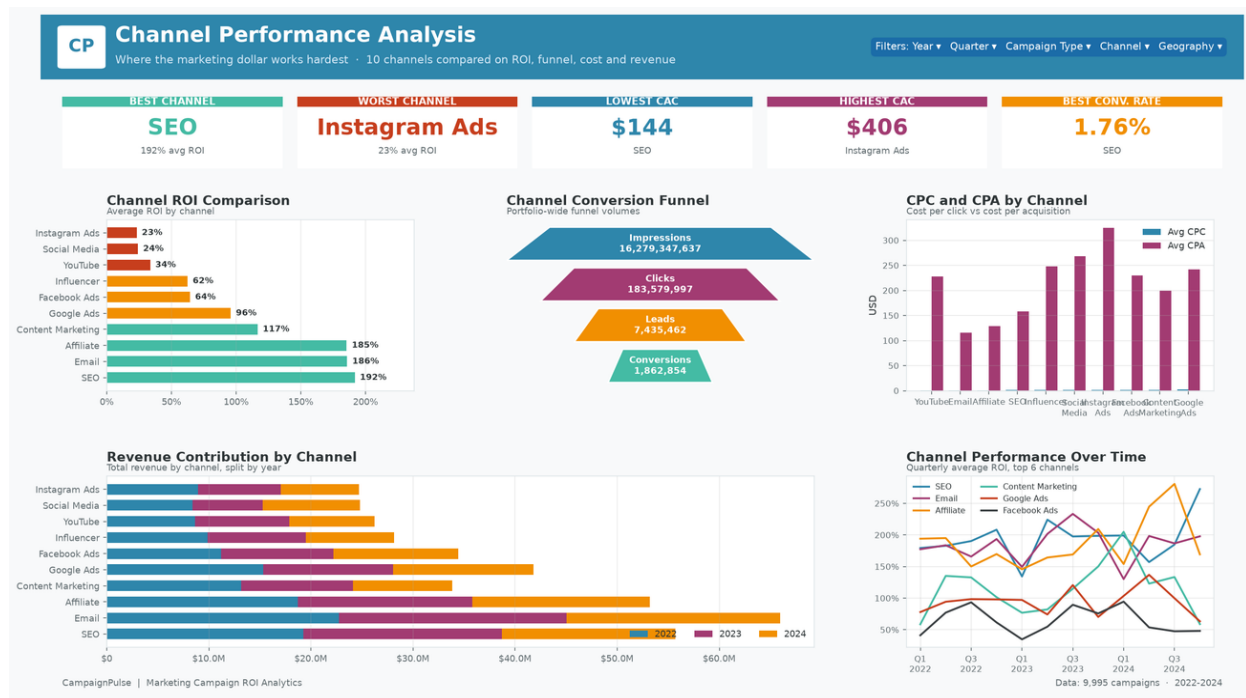
Four Tableau dashboards translate the analysis into an interactive decision tool. Each carries Year, Quarter, Campaign Type, Channel and Geography filters, interactive tooltips and the CampaignPulse colour theme. The previews below are rendered from the same cleaned dataset.

8.1 Executive Overview



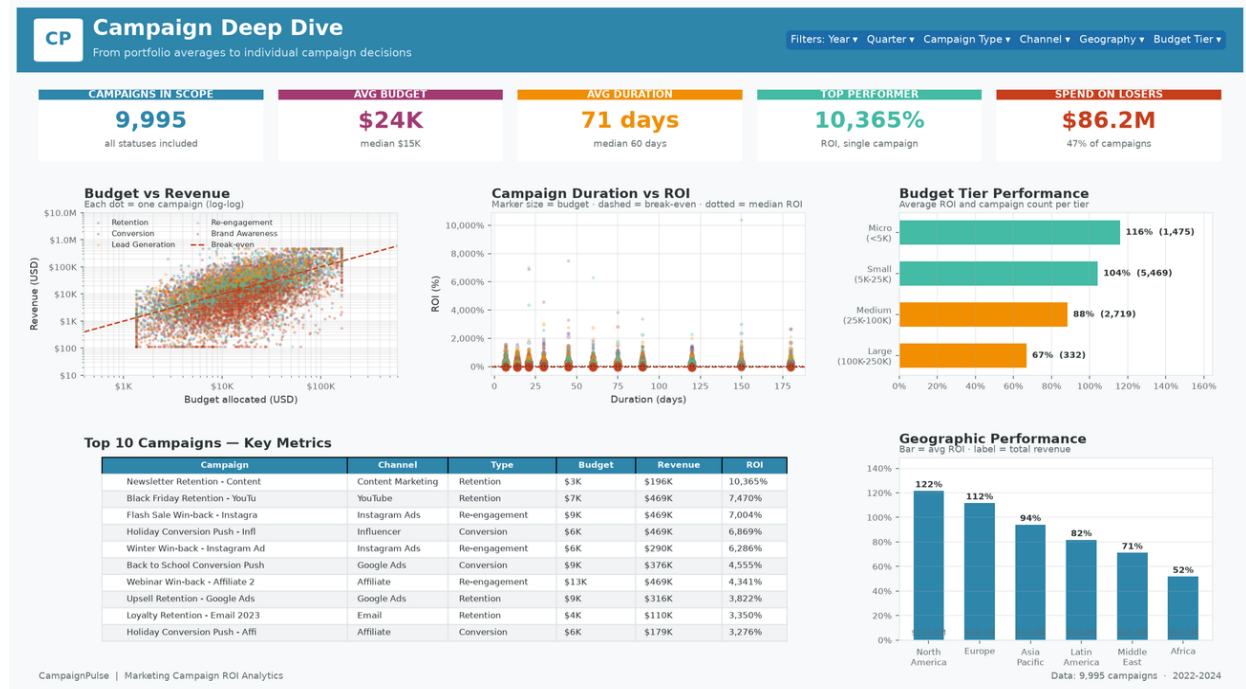
KPI cards for revenue, spend, ROI, campaign count and CAC; monthly revenue versus spend; ROI by campaign type; status distribution; profitability split; ROI by region.

8.2 Channel Performance Analysis



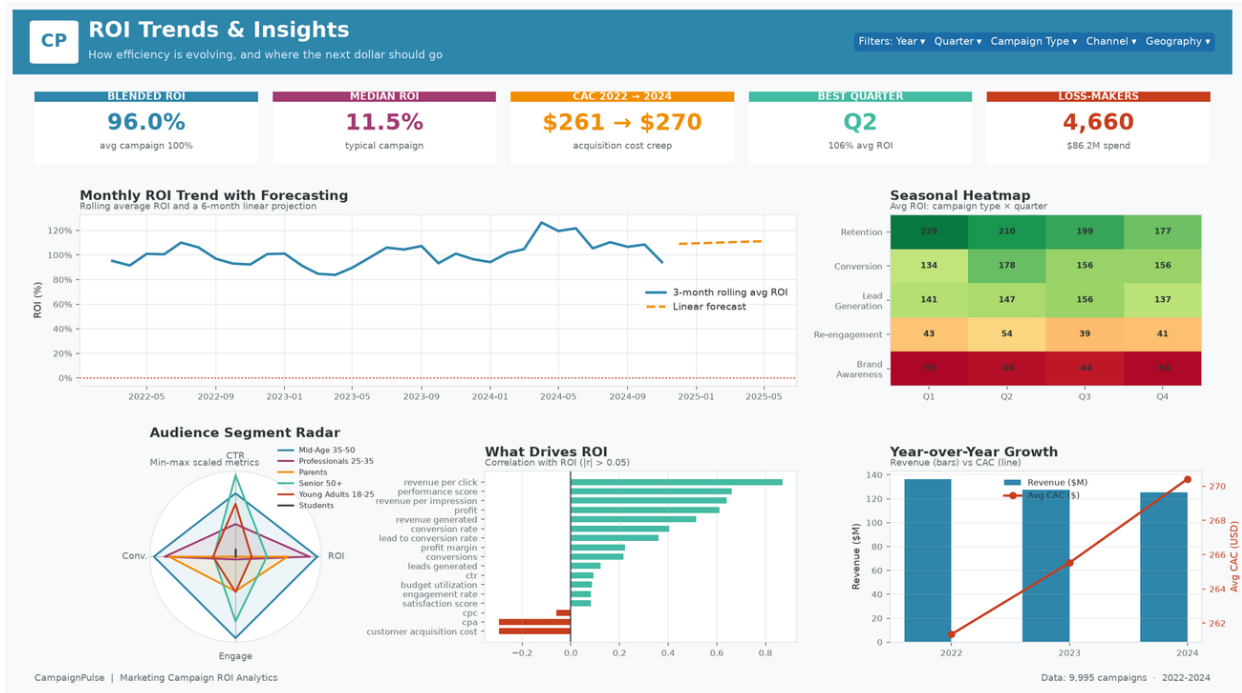
Channel ROI comparison, conversion funnel, CPC/CPA dual axis, revenue contribution treemap and channel performance over time.

8.3 Campaign Deep Dive



Budget-versus-revenue scatter, duration versus ROI, the top-10 campaign table, budget-tier performance and geographic performance.

8.4 ROI Trends & Insights



Monthly ROI trend with forecasting, seasonal heatmap, audience-segment radar, correlation analysis and year-over-year growth.

9. Key Findings

1. Retention is the best campaign type

203.9% average ROI — double the 100.4% portfolio average and the best type in all four quarters.

2. Brand Awareness destroys value

–50.8% average ROI, negative in every quarter, only 10.8% of campaigns profitable.

3. Owned and earned media outperform paid social

SEO 191.7%, Email 185.8%, Affiliate 185.4% against Instagram 23.1% and Social Media 24.1%.

4. CAC is the master variable

The strongest negative correlate of ROI ($r = -0.30$). Email acquires a customer for \$144, Instagram for \$406.

5. Nearly half of all campaigns lose money

46.6% negative ROI; median ROI only 11.5% against a mean of 100.4%.

6. Diminishing returns to budget are real

Revenue elasticity of budget ≈ 0.70 ; ROI falls from 115.9% (Micro) to 66.8% (Large).

7. Top performers are small and precise

All ten highest-ROI campaigns are \$2.8K–\$13.3K Retention or Conversion campaigns.

8. Mid-Age and Professional segments are the sweet spot

167.5% and 152.5% ROI with the lowest CAC; Students and Young Adults are weakest with the highest CAC.

9. Campaign duration is not a performance lever

Correlation with ROI ≈ 0.00 across every duration bucket.

10. Efficiency has been flat for three years

Blended ROI 98.8% $\rightarrow$ 92.1% $\rightarrow$ 97.1% while CAC rose from \$261 to \$270.

11. North America and Europe carry the portfolio

\$232M of \$389M revenue at 121.6% and 111.7% ROI; Africa is weakest at 51.8%.

12. Quarter effects are small relative to type effects

Q2 106.3% versus Q1 94.7% — a 12-point spread, against a 250-point spread between the best and worst campaign types.

10. Business Recommendations

#	Recommendation	Priority	Detail
1	Rebalance the campaign-type mix toward Retention	High	Shifting 15% of Brand Awareness budget into Retention at current average ROIs would
2	Cap or re-purpose Brand Awareness	High	It currently spends \$39.9M and returns \$21.2M. Either cap it as a defined brand-building lin
3	Shift paid-social budget into SEO, Email and Affiliate	High	These three channels combine the highest ROI with the lowest CAC. Reduce Instagram an
4	Manage to CAC, not to CPC or CTR	High	CPC and CTR are uncorrelated or negatively correlated with ROI, while CAC is the stronge
5	Concentrate spend on Mid-Age 35-50 and Professional Students and Young Adults	Medium	Reallocate 35% of the budget, or rebuild the post-click experience for those
6	Release incremental budget only to Mothers	Medium	With elasticity ≈ 0.70 , incremental budget should be released in small increments and only
7	Triage the loss-making cohort	Medium	Diagnose whether the cause is targeting, creative, offer or channel, then pause, fix or reallo
8	Adopt median-based reporting in dashboards	Low	Median ROI is 11.5% against a mean of 100.4%. Average-based dashboards flatter perform
9	Launch a productivity programme	Medium	Revenue is flat while CAC has risen ~3.5% since 2022. The portfolio needs efficiency gains
10	Institutionalise the data-quality gates	Low	Keep the validation rules from the cleaning notebook in production so bad data can never a

11. Conclusion

CampaignPulse turned a 10,018-row raw extract into a validated, 9,995-row analytical dataset and a complete decision-support package. The analysis shows a portfolio that looks healthy in aggregate — a 96.0% blended ROI — but is highly uneven underneath: nearly half of all campaigns lose money, one campaign type destroys value outright, and the channels with the largest budgets are not the channels with the best returns.

The clearest conclusion is that **objective and cost discipline matter far more than scale**. Retention and Conversion campaigns, small budgets, owned and earned channels, and mid-career audiences consistently outperform. Meanwhile budget size, impressions, clicks and campaign duration explain almost nothing about ROI.

Acting on the recommendations in section 10 would address the largest profit pool in the portfolio — the \$86M spent on campaigns that never pay back — without requiring a single additional dollar of budget.

Limitations

- Revenue attribution is taken as given; no multi-touch attribution model is applied.
- Customer acquisition cost is modelled as direct CPA plus a 25% overhead allocation.
- Outliers are capped rather than removed, so average ROI is a slightly conservative view of the break-out campaigns.
- The dataset is synthetic, generated to be statistically realistic for portfolio and demonstration purposes.

12. Appendix

A. Repository contents

B. Validation summary

C. References and resources

- pandas documentation — <https://pandas.pydata.org/docs/>
- matplotlib documentation — <https://matplotlib.org/stable/contents.html>
- MySQL 8.0 reference manual — <https://dev.mysql.com/doc/refman/8.0/en/>
- Tableau Desktop help — <https://help.tableau.com/current/pro/desktop/en-us/>
- Project repository — <https://github.com/AmitKumarJha-ds/marketing-campaign-roi-analytics>

D. Licence

This project is released under the MIT Licence. See the LICENSE file in the repository root for the full text.